Sample MCQ Questions for Test 1

1. A student designs a cipher replacing A with Z and Z with A. Which classic cipher model does this describe?
   - Atbash Cipher ✓
   - ROT13 Cipher
   - Keyword Cipher
   - Vigenère Cipher

2. A bank encrypts large volumes of data using a single secret key shared between sender and receiver. Which encryption model is used?
   - Symmetric encryption ✓
   - Asymmetric encryption
   - Stream hashing
   - Public key exchange

3. A student designs a random generator where output depends on previous output values. What issue might this create?
   - Non-determinism
   - Reduced entropy ✓
   - Infinite recursion
   - Excess keyspace

4. Which asymmetric algorithm provides equivalent security to RSA with smaller key sizes, making it ideal for mobile devices?
   - DSA
   - ECC ✓
   - ElGamal
   - Diffie–Hellman

5. In Cipher Feedback (CFB) mode, what happens if one bit in ciphertext changes during transmission?
   - All plaintext is corrupted
   - Only one plaintext bit changes
   - Two blocks are affected ✓
   - No change occurs

6. Why can’t hash functions alone provide message authentication?
    - They lack encryption
    - They are too slow
    - They don't use a secret key ✓
    - They produce variable-length outputs

7. Why is “security by obscurity” considered inadequate for modern cryptography?
    - It conceals the algorithm instead of securing the key ✓
    - It makes key distribution impossible
    - It depends on weak mathematics
    - It limits algorithmic flexibility

8. What distinguishes a keyed hash (HMAC) from a basic hash function?
    - It uses a key and two hashing rounds ✓
    - It has a longer digest
    - It is reversible
    - It eliminates collisions